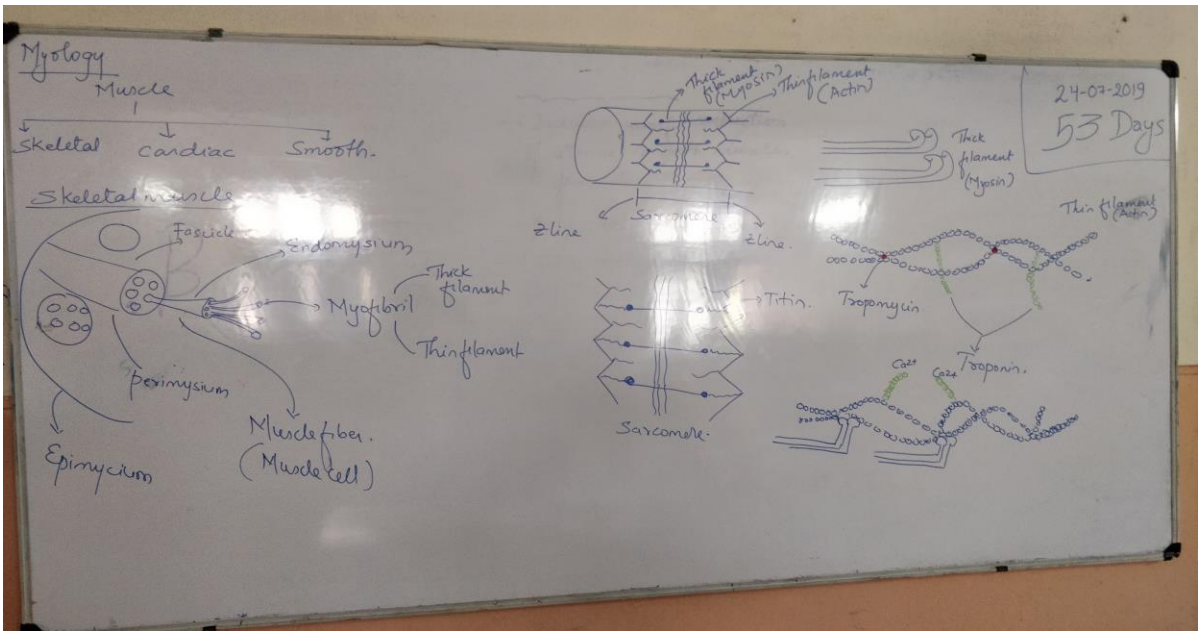
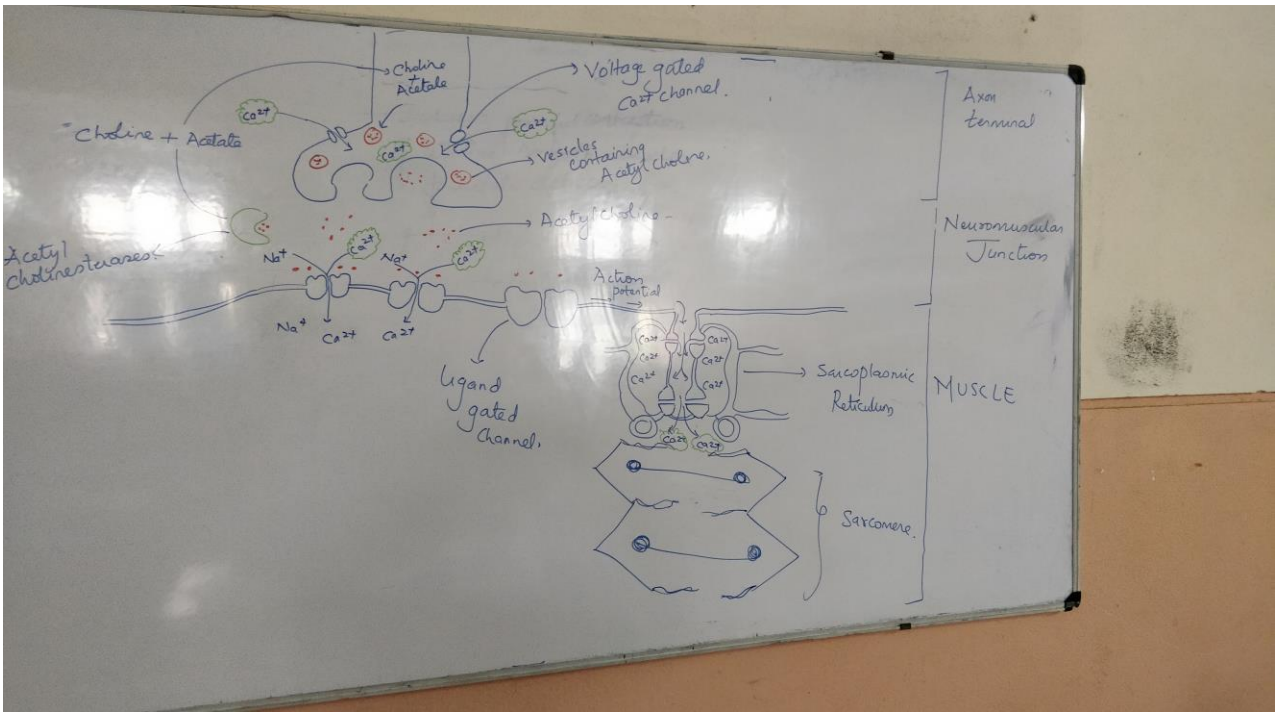
Physiology of Muscle Contraction

Skeletal muscles contract according to the *sliding filament model*:

1. An action potential originating in the CNS reaches an alpha motor neuron, which then transmits an action potential down its own axon.
2. The action potential propagates by activating sodium dependent channels along the axon toward the synaptic cleft. Eventually, the action potential reaches the motor neuron terminal and causes a calcium influx through the calcium-dependent channels.
3. The calcium influx causes vesicles containing the neurotransmitter acetylcholine to fuse with the plasma membrane, releasing acetylcholine out into the extracellular space between the motor neuron terminal and the motor end plate of the skeletal muscle fiber.
4. The acetylcholine diffuses across the synapse and binds to and activates nicotinic acetylcholine receptor on the motor end plate of the muscle cell. Activation of the nicotinic receptor opens its intrinsic sodium/potassium channel, causing sodium to rush in and potassium to trickle out. Because the channel is more permeable to sodium, the muscle fiber membrane becomes more positively charged, triggering an action potential.
5. The action potential spreads through the muscle fiber's network of T-tubules, depolarizing the inner portion of the muscle fiber.
6. The depolarization activates L-type voltage-dependent calcium channels (dihydropyridine receptors) in the T tubule membrane, which are in close proximity to calcium-release channels (ryanodine receptors) in the adjacent sarcoplasmic reticulum.
7. Activated voltage-gated calcium channels physically interact with calcium-release channels to activate them, causing the sarcoplasmic reticulum to release calcium.
8. The calcium binds to the troponin C present on the actin-containing thin filaments of the myofibrils. The troponin then allosterically modulates the tropomyosin. Normally the tropomyosin sterically obstructs binding sites for myosin on the thin filament; once calcium binds to the troponin C and causes an allosteric change in the troponin protein, troponin T allows tropomyosin to move, unblocking the binding sites.
9. Myosin (which has ADP and inorganic phosphate bound to its nucleotide binding pocket and is in a ready state) binds to the newly uncovered binding sites on the thin filament (binding to the thin filament is very tightly coupled to the release of inorganic phosphate). Myosin is now bound to actin in the strong binding state. The release of ADP and inorganic phosphate are tightly coupled to the power stroke (actin acts as a cofactor in the release of inorganic phosphate, expediting the release). This will pull the Z-bands towards each other, thus shortening the sarcomere and the I-band.
10. ATP binds myosin, allowing it to release actin and be in the weak binding state (a lack of ATP makes this step impossible, resulting in the rigor state characteristic of rigor mortis). The myosin then hydrolyzes the ATP and uses the energy to move into the "cocked back" conformation. In general, evidence (predicted and *in vivo*) indicates that each skeletal muscle myosin head moves 10-12 nm each power stroke, however there is also evidence (*in vitro*) of variations (smaller and larger) that appear specific to the myosin isoform
11. Steps 9 and 10 repeat as long as ATP is available, and calcium is present on thin filament.
12. While the above steps are occurring, calcium is actively pumped back into the sarcoplasmic reticulum. When calcium is no longer present on the thin filament, the tropomyosin changes conformation back to its previous state so as to block the binding sites again. The myosin ceases binding to the thin filament, and the contractions cease.


Myology
Muscle
Skeletal
cardiac
Smooth.
Skeletal muscle
Fascicle
Endomysium
Myofibril
Thick filament
Thin filament
Perimysium
Epimysium
Muscle fiber.
(Muscle cell)
Thick filament (Myosin)
Thin filament (Actin)
Z line
Sarcomere
Z line.
Titin.
Sarcomere.
Thick filament (Myosin)
Thin filament (Actin)
Tropomyosin
Troponin.
24-07-2019
53 Days


Choline + Acetate
Choline + Acetate
Voltage gated Ca2+ channel.
Vesicles containing Acetyl Choline.
Acetyl Choline
Acetyl cholinesterase
Na+
Ca2+
Action potential
Ligand gated channel.
Sarcoplasmic Reticulum
Sarcomere.
Axon terminal
Neuromuscular Junction
MUSCLE